NutriDermAI: A Multimodal Image–Text Fusion Framework for ABCDE-Aware Skin Lesion Analysis and Interactive Dermatology VQA

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***Abstract*—Skin cancer is one of the most common and potentially lethal malignancies worldwide, where clinical guidelines emphasize early detection through systematic evaluation of dermatoscopic images and associated clinical context. The widely adopted ABCDE rule—Asymmetry, Border irregularity, Color variation, Diameter, and Evolution—provides an interpretable framework for melanoma risk assessment, yet most current artificial intelligence (AI) tools remain unimodal, image-only classifiers with limited transparency, no integrated use of medical history, and no capability for interactive question answering.**

This paper proposes *NutriDermAI*, a multimodal dermatology assistant that unifies (i) skin lesion segmentation, (ii) lesion region-of-interest (ROI) extraction, (iii) hierarchical lesion classification with image-based ABCD computation, (iv) medical text understanding for deriving "E" and clinical context, (v) an Image–Text Fusion (IT-Fusion) layer for calibrated ABCDE risk scoring, and (vi) a Visual Question Answering (VQA) module for interactive, patient-centric and clinician-centric dialogue. The system is implemented as a nine-stage pipeline, covering dataset preprocessing, model training, multimodal fusion, conversational reasoning, and deployment. We describe the architecture, stage-wise design, target datasets, planned models, and evaluation protocol, positioning NutriDermAI against recent advances in dermatology foundation models, multimodal medical AI, and medical VQA. The work is framed as a proposal for a deployable, explainable, and safety-aware dermatology support system suitable for conference-level publication and future clinical validation.

***Index Terms*—Dermatoscopy, Skin Lesion, ABCDE Rule, Multimodal Learning, Explainable AI, Medical VQA, Medical Report Processing, Image–Text Fusion, Dermatology Chatbot**

I. Introduction

Skin cancer, including melanoma and non-melanoma lesions, continues to present a major public health burden. Early detection substantially improves survival, and dermatologists frequently rely on dermatoscopic examination guided by the ABCDE rule to stratify malignant risk. Classical ABCD/ABCDE-based expert systems and rule-driven scoring approaches have demonstrated that structured feature assessment (e.g., asymmetry, border irregularity, color variegation, diameter, evolution) can approximate dermatologists' diagnostic reasoning [1]–[5].

Concurrently, deep learning has achieved high performance on dermatoscopic segmentation and classification tasks using U-Net variants, hybrid CNN–Transformer designs, and large-scale dermatology datasets such as HAM10000, ISIC 2016–2024, BCN20000, and MiLK-10K [6]–[17]. More recently, dermatology foundation models and multimodal vision–language models (VLMs) such as PanDerm, MM-Skin, and Derm1M have shown how large-scale, expert-curated multimodal supervision improves zero-shot recognition and text grounding in dermatology [18]–[22].

In parallel, medical natural language processing (NLP) and medical VQA have evolved rapidly, leveraging clinical corpora like MIMIC-III/IV and specialized architectures such as BioBERT, medical multimodal large language models (MM-LLMs), and reinforcement learning (RL)-trained VQA systems [23]–[33]. Dermatology-specific VQA benchmarks and chat-based assistants are emerging, yet they typically lack end-to-end ABCDE computation and structured fusion of imaging with clinical text [34]–[44].

Motivation. Existing dermatology AI solutions can (i) segment lesions, (ii) classify lesions into benign/malignant or multi-class taxonomies, or (iii) support generic multimodal question answering. However, there is still a gap for a unified dermatology assistant that:

- Computes explicit ABCDE values in a manner aligned with clinical ABCDE literature and scoring rules;
- Integrates dermatoscopic images with patient symptoms, history, and laboratory or biopsy reports;
- Produces calibrated risk scores and structured JSON explanations usable by downstream applications;
- Supports safe, context-aware VQA with explicit rationales grounded in ABCDE features and clinical evidence.

Contribution. This paper proposes *NutriDermAI*, a nine-stage multimodal ABCDE-aware pipeline that:

1) Preprocesses image and text datasets for training and deployment (Stage 1).
2) Performs skin lesion segmentation (SLS) and lesion ROI cropping (SLRC) (Stages 2–3).
3) Learns hierarchical lesion classification and extracts image-based ABCD metrics (Stage 4).

4) Processes medical history and reports to derive evolution (E) and clinical context (Stage 5).
5) Fuses all evidence in an IT-Fusion layer to compute final ABCDE, risk scores, and explanations (Stage 6).
6) Uses a dermatology-aware VQA module for interactive dialogue with patients and clinicians (Stage 7).
7) Integrates frontend, backend, and deployment components with appropriate testing (Stages 8–9).

The paper is structured as a proposal-style research article suitable for conferences, outlining problem definition, related work, architecture, datasets, implementation plan, evaluation methodology, and ethical considerations.

II. Background and Related Work

A. ABCDE Rule and Classical Dermatology Decision Support

The ABCD/ABCDE rule offers a structured approach to melanoma risk assessment by quantifying asymmetry, border irregularity, color variation, diameter, and evolution over time. Classical expert systems and analytic pipelines compute scores from dermatoscopic images and combine them into total dermatoscopic scores (TDS) [1]–[3]. Recent work has shown that generative models and AI systems that explicitly incorporate ABCDE analysis can match or approach dermatologist-level decision-making and provide interpretable outputs [4], [5]. Hybrid CNN+ABCD models also demonstrate that explicit ABCD descriptors can boost classification performance when coupled with deep features [45].

B. Skin Lesion Segmentation

Lesion segmentation is a prerequisite for many downstream tasks, including ABCDE feature computation and ROI cropping. U-Net and its numerous variants remain dominant in medical image segmentation [6], [46]. For skin lesions, dermatology-specialized architectures such as ARC-UNet, ScaleFusionNet, ESC-UNet, MRP-UNet, Meta-UNet, BiADATU-Net, MUCM-Net, and dual CNN–ViT hybrids have improved boundary precision, small-lesion sensitivity, and robustness by leveraging residual connections, deformable attention, multiscale inputs, and metadata [7]–[13], [47]. Multiscale Swin Transformer-based models further improve segmentation across diverse medical imaging domains [48], [49].

C. Lesion Classification and Dermatology Foundation Models

Large dermatoscopic datasets—including ISIC 2016–2024 challenges, HAM10000, BCN20000, and MiLK-10K—have enabled high-performing lesion classifiers and foundation models [14]–[17]. Efficient transformer-based backbones such as EfficientFormerV2 provide competitive accuracy with reduced memory footprint [50], making them suitable for multi-stage pipelines. Recent dermatology foundation models like PanDerm, MM-Skin, and Derm1M-scale vision–language datasets align dermatoscopic images with rich textual descriptions and hierarchical ontologies [18]–[21]. These works motivate the use of shared multimodal encoders for segmentation, classification, and fusion.

D. Clinical Text Understanding and Medical NLP

Clinical narratives, pathology reports, and lab summaries contain crucial contextual information—including symptom evolution and risk factors. BioBERT and related biomedical-language encoders have improved named entity recognition and classification over biomedical literature and clinical notes [23]. Large ICU corpora such as MIMIC-III and MIMIC-IV Note provide rich free-text and structured data for training and evaluation [24], [25]. Systematic reviews of clinical text data highlight both opportunities and pitfalls for machine learning on noisy, heterogeneous EHR text [26]. These insights inform our MedProc stage, where symptom descriptions, timelines, and lab values are normalized into structured features.

E. Medical Multimodal LLMs and VQA

Medical VQA research has produced datasets such as VQA-RAD, PathVQA, TM-PathVQA, HEALMQA, and dermatology-specific DermaVQA, which cover diverse modalities and question types [34]–[37], [51]. Model architectures range from BLIP-style multimodal encoders (BioMedBLIP) to specialized VQA fusion layers with semantic attention [27], [28]. Dynamic prompt adaptation and RL-based optimization have been proposed to reduce hallucinations and improve reasoning in medical VQA [29]–[32], [52].

Dermatology-focused works demonstrate joint learning of segmentation and VQA, CLIPSeg-based multimodal fusion, and knowledge-enhanced ensembles [38], [39], [53]–[55]. NutriDermAI is aligned with these directions but adds explicit ABCDE reasoning, rule-based cross-checks, and a unified JSON explanation layer.

F. Dermatology Chatbots and Conversational Diagnostic AI

Several works have explored dermatology chatbots combining CNN-based lesion analysis with rule-based triage logic [40], [41]. Broader diagnostic LLMs like AMIE evaluate conversational diagnostic quality across multiple specialties and emphasize safety, uncertainty communication, and escalation strategies [43]. Reviews of ChatGPT-style models in dermatology and multimodal LLMs in medicine further highlight the need for guardrails, bias mitigation, and rigorous evaluation [33], [42]. Studies on conversational agents in dermatology show improved patient engagement but also underline responsibility and ethical concerns [44]. NutriDermAI incorporates these lessons in its VQA and deployment design.

G. Gap and Design Requirements

From this literature, key gaps emerge:

- ABCDE-aware systems that compute all five criteria in a clinically consistent, data-driven manner remain rare.
- Integration of dermatoscopic images with detailed medical history and laboratory reports is limited, especially in dermatology.
- Medical VQA systems often lack explicit links between answers and structured evidence such as ABCDE scores or rule-based checks.

- End-to-end dermatology assistants spanning segmentation, classification, multimodal fusion, VQA, and production deployment are not widely documented.

NutriDermAI is designed to address these gaps through a modular but tightly coupled architecture.

III. System Overview

NutriDermAI is implemented as a nine-stage pipeline (Fig. 1):

- Stage 1: Preprocessing of datasets and raw inputs (images and text).
- Stage 2: Skin Lesion Segmentation (SLS).
- Stage 3: Skin Lesion ROI Cropping (SLRC).
- Stage 4: Hierarchical Classification + raw ABCD extraction (HC).
- Stage 5: MedProc – medical information processing and Evolution (E).
- Stage 6: Image + Text Fusion (IT-Fusion) for final ABCDE and risk.
- Stage 7: Visual Question Answering (VQA).
- Stage 8: Frontend–Backend and API layer.
- Stage 9: Testing, monitoring, and deployment.

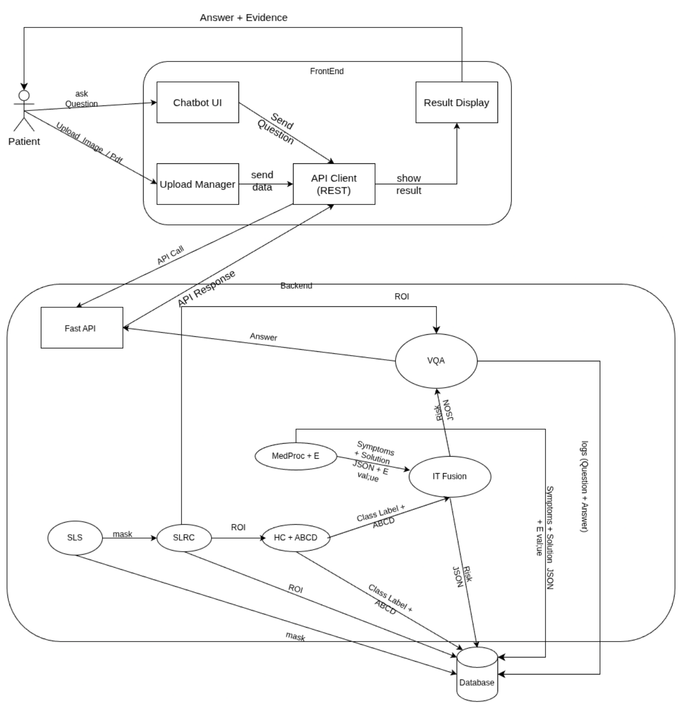

Fig. 1: High-level NutriDermAI architecture: staged pipeline from raw patient inputs to ABCDE-aware risk assessment and VQA.

All stages exchange structured artifacts (e.g., binary masks, ROIs, ABCD scores, symptom vectors, unified JSON) to support transparency, modular retraining, and debugging.

IV. Stage-Wise Methodology

A. Stage 1: Dataset and Input Preprocessing

Stage 1 prepares both training datasets and live user inputs for downstream stages.

preprocessing.png

Fig. 2: Preprocessing stage: unifying image and text inputs into standardized training-ready and inference-ready formats.

For image data, we perform:

- Resolution standardization (e.g., resizing to 512×512 or model-compatible sizes).
- Pixel-value normalization and optional contrast enhancement.
- Alignment of mask–image pairs for segmentation datasets.
- Removal of artefacts (e.g., hair, ruler marks) when required.

For textual data (medical history, symptoms, reports), we apply:

- OCR for scanned PDFs or images of reports.
- Text normalization (lowercasing, de-noising, sentence segmentation).
- Medical term tokenization and preliminary keyword tagging.

B. Stage 2: Skin Lesion Segmentation (SLS)

Given a full skin lesion image, the SLS stage estimates a binary lesion mask.

We plan to use ARC-UNet as the primary backbone, taking advantage of residual connections and attention modules tailored to dermatoscopic images [7]. Alternative or complementary models such as ScaleFusionNet, ESC-UNet, MRP-UNet, and Swin-based segmentation are considered for ablations and ensemble strategies [8]–[10], [48].

Outputs include the binary mask, contour geometry, area, and mask quality indicators. These outputs are critical for

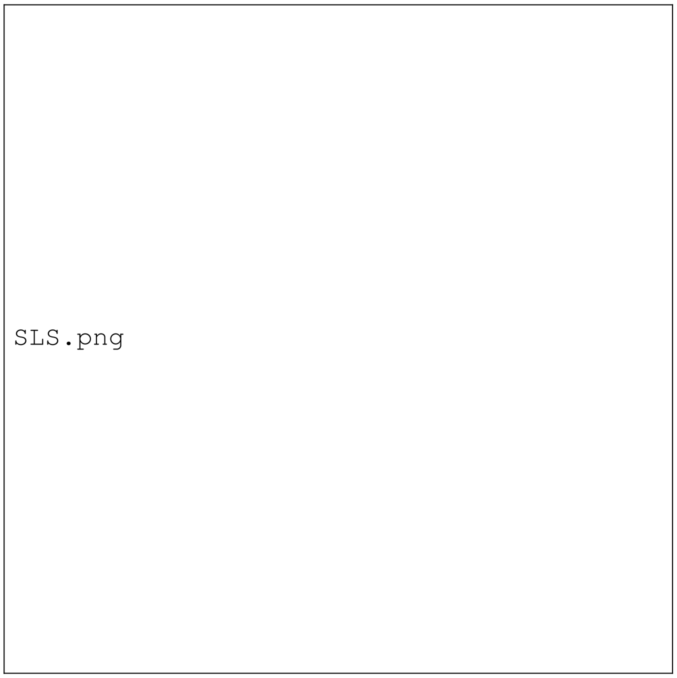

Fig. 3: Stage 2 (SLS): segmentation of lesion area and extraction of geometric mask descriptors.

ABCD computation (e.g., asymmetry, border metrics) and SLRC.

C. Stage 3: Skin Lesion ROI Cropping (SLRC)

Stage 3 uses the SLS mask to crop a color ROI that tightly covers the lesion.

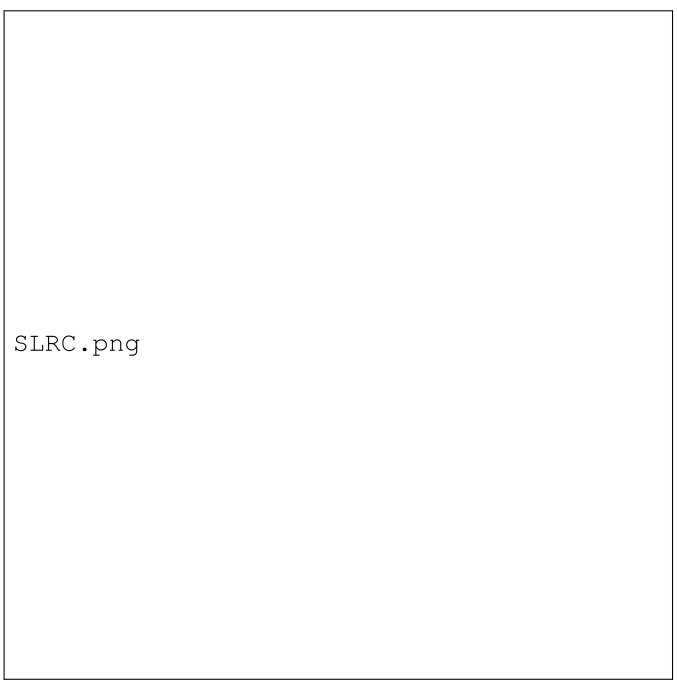

Fig. 4: Stage 3 (SLRC): mask-aware ROI cropping and normalization for classification.

The mask is aligned with the original image; a bounding box around the largest connected component is computed, expanded with a safety margin, and cropped. The ROI is then resized to a canonical size (e.g., 224×224) for classification. Metadata such as bounding boxes, scaling factors, and mask coverage ratio are preserved for later consistency checks.

D. Stage 4: Hierarchical Classification and Raw ABCD Extraction (HC)

In Stage 4, the ROI is passed to a hierarchical classifier built on a lightweight Vision Transformer backbone such as EfficientFormerV2 [50].

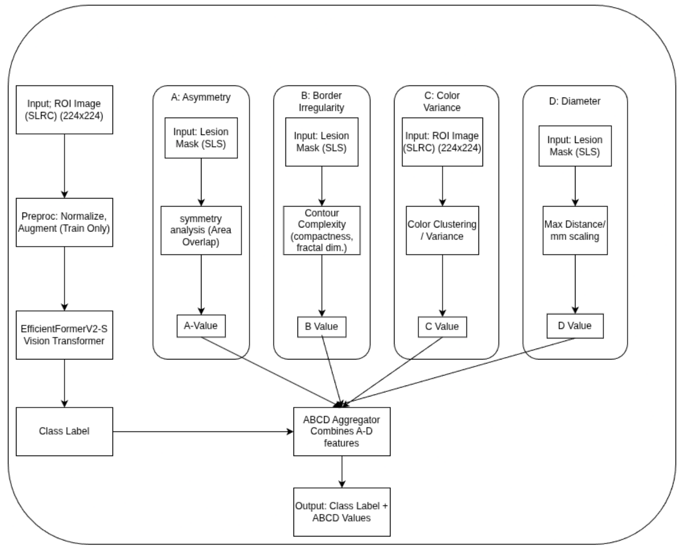

Fig. 5: Stage 4 (HC): hierarchical classification and computation of raw ABCD metrics from ROI and segmentation mask.

The classifier outputs:

- Level 1: benign vs. malignant probabilities.
- Level 2: multi-class predictions (e.g., melanoma, nevus, seborrheic keratosis, other lesion types) when supported by training data.

Simultaneously, the SLS mask and ROI are used to compute image-based ABCD metrics:

- **Asymmetry (A):** shape imbalance computed by comparing overlaps between mirrored halves of the lesion mask along orthogonal axes [1], [2].
- **Border Irregularity (B):** contour complexity, perimeter-to-area ratios, and fractal-like indices derived from the binary mask [11], [12].
- **Color Variation (C):** number and dispersion of color clusters (e.g., via k-means in CIELAB space) within the ROI [45].
- **Diameter (D):** maximum Euclidean distance within the lesion mask, scaled by image resolution when available, to approximate millimeter measurements [3].

The HC stage returns classification logits, ABCD values, confidence estimates, and provenance tags (model version, training configuration).

E. Stage 5: MedProc – Medical Information Processing

Stage 5 processes multimodal clinical text, including:

- Chat responses describing symptoms (itching, bleeding, growth rate).
- Uploaded biopsy or blood-test reports (e.g., LDH levels).
- Previous lesion images and historical notes.

A BioBERT-based encoder is fine-tuned for clinical entity recognition and relation extraction, leveraging MIMIC-III and MIMIC-IV notes as foundational corpora [23]–[25]. Rule-based logic built on ABCDE literature maps extracted entities to:

- Evolution (E)—capturing temporal changes in size, color, bleeding, itching, etc.
- Risk factors—e.g., sun exposure, family history, immunosuppression.
- Non-dermatological symptoms to support general physician decision support.

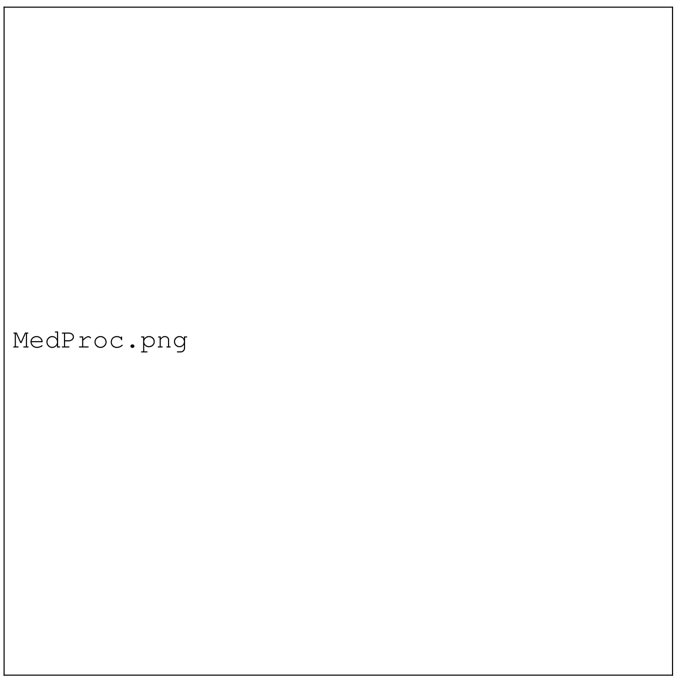

Fig. 6: Stage 5 (MedProc): extraction of Evolution (E) and clinical features from patient history and reports.

Outputs are structured as JSON-formatted symptom vectors, Evolution scores, numerical values (e.g., lab results), and textual rationales.

F. Stage 6: IT-Fusion – Final ABCDE and Risk Computation

Stage 6 combines image-based ABCD metrics, text-derived Evolution (E), classification logits, segmentation metadata, and symptom summaries.

Inspired by multimodal transformer frameworks such as MARIA and RAG-style fusion [56]–[58], we design an IT-Fusion component that:

1) Normalizes and encodes numerical features (ABCD, E, logits, geometry).

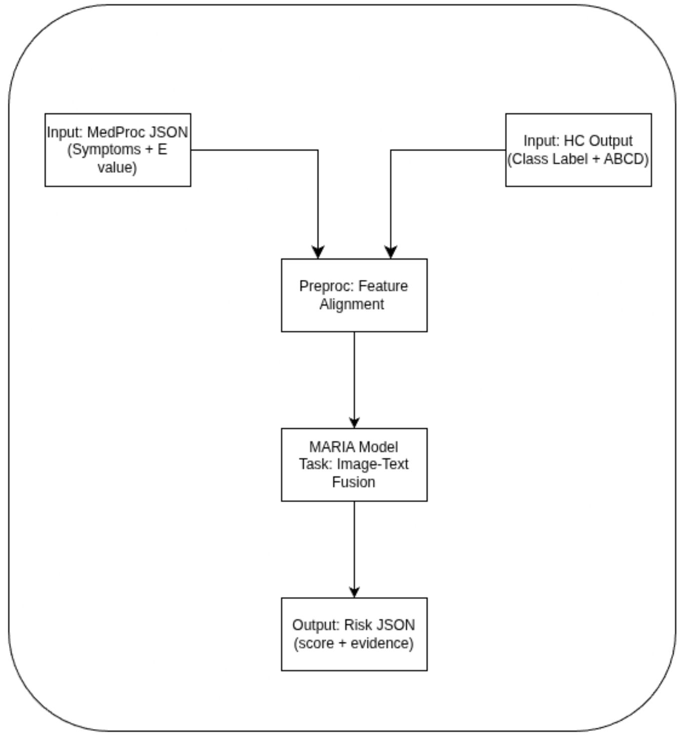

Fig. 7: Stage 6 (IT-Fusion): multimodal aggregation of ABCD, Evolution, classification outputs, and clinical features into a unified ABCDE-aware risk estimate.

2) Applies cross-attention mechanisms between image features, text embeddings, and structured features.
3) Performs rule-based sanity checks (e.g., verifying consistent diameter scales, flagging conflicting evolution information).
4) Outputs calibrated final ABCDE scores, a risk probability, and a final lesion class.

The stage returns a unified JSON object containing:

- Final ABCDE values.
- Final benign/malignant and multi-class decisions.
- Risk score and calibration metadata.
- Evidence trace listing which features and stages contributed to the decision.
- Flags for missing data or cases requiring dermatologist review.

G. Stage 7: Visual Question Answering (VQA)

The VQA module consumes:

- The lesion ROI image and optionally the full image.
- The IT-Fusion unified JSON.
- The conversation history and the current natural-language question.

As a baseline, we consider adapting R-LLaVA or similar Med-VQA models that explicitly leverage ROI guidance for improved answers [27], [38], [53]. Additional enhancements from MedVLM-R1, DSAP, MedThink, and Med-GRIM provide candidate RL and prompting strategies to balance answer

VQA.png

Fig. 8: Stage 7 (VQA): dermatology-focused vision–language model answering patient and clinician questions using the unified JSON and lesion imagery.

accuracy, calibration, and reduced hallucination [29], [30], [32], [52], [59], [60].

The model is trained and evaluated on a combination of general medical VQA datasets (PathVQA, VQA-RAD, HEALMQA, TM-PathVQA) and dermatology-specific DermaVQA questions [34]–[37], [51]. Answers are generated together with short rationales referencing ABCDE values and explicit evidence items in the JSON.

H. Stage 8: Frontend–Backend and API Layer

Stage 8 exposes the system as a chat-based web application with:

- A React-based frontend for uploading lesion images, attaching reports, and entering questions.
- A FastAPI backend orchestrating SLS, SLRC, HC, MedProc, IT-Fusion, and VQA as modular microservices.
- Secure storage and retrieval of intermediate artifacts (e.g., masks, JSON, logs) for auditability.

Design considerations are informed by dermatology chatbot case studies and conversational diagnostic LLM frameworks that highlight UX, disclaimers, and escalation pathways [40], [41], [43], [44].

I. Stage 9: Testing, Monitoring, and Deployment

Stage 9 focuses on:

- Unit and integration tests for all stages.
- Offline evaluation on held-out datasets (see Section V).
- Continuous monitoring for drift, error patterns, and user feedback.
- Deployment to a controlled environment (e.g., internal research server) with appropriate access controls and logging.

V. Datasets and Experimental Design

A. Stage-Wise Dataset Allocation

Table I summarizes the planned datasets per stage, drawing on standard dermatoscopic collections and clinical text corpora [14]–[17], [20], [24], [25], [34]–[36].

B. Data Curation and Preprocessing

Because datasets originate from different centers with varying acquisition protocols, we will:

- Harmonize label taxonomies (e.g., mapping to a common ontology such as Derm1M's hierarchy [20]).
- Standardize image resolutions and color profiles.
- Create clean splits where patient-level leakage is minimized.
- For VQA, generate dermatology-specific question–answer pairs from Derm1M captions and literature where necessary, following methods that synthesize QA from medical texts [54].

VI. Implementation Plan

A. Model Choices

For each stage, candidate models are selected based on performance and feasibility with GPUs of around 12 GB VRAM:

- **SLS:** ARC-UNet as primary segmentation backbone, with optional comparison to ScaleFusionNet, ESC-UNet, MRP-UNet, Meta-UNet, BiADATU-Net, MUCM-Net, and CNN–ViT hybrids [7]–[13], [47].
- **HC:** EfficientFormerV2 backbone for hierarchical classification and ABCD-based feature fusion [45], [50].
- **MedProc:** BioBERT-based encoder fine-tuned on MIMIC-III/IV notes, with rule-based ABCDE logic grounded in dermatology literature [1]–[3], [23].
- **IT-Fusion:** MARIA-style multimodal transformer with attention-based modality fusion, extended with rule-based validators [56]–[58].
- **VQA:** An R-LLaVA-style ROI-aware Med-VQA model combined with knowledge-enhanced ensembles and RL-based answer reranking [27], [29], [30], [32], [38], [52], [53], [59], [60].

B. Training Strategy

Each stage is trained separately, followed by joint fine-tuning:

- Pre-train SLS and HC using standard segmentation and classification losses (Dice, cross-entropy, focal loss).
- Train MedProc on clinical NER and relation tasks; validate on manually curated dermatology notes.
- Train IT-Fusion on multimodal samples with ground-truth ABCDE and labels, using calibration-aware losses (e.g., Brier score).
- Train VQA on a mix of general Med-VQA and dermatology-specific VQA datasets, with additional reinforcement signals for answer correctness and format.

TABLE I: Planned datasets for each stage in NutriDermAI. Counts are approximate and may be refined during implementation.

Stage	Task	Dataset(s)	Train	
SLS	Segmentation	ISIC 2016, ISIC 2017 – Segmentation tasks	2,900	
SLRC	ROI Cropping	ROIs derived from SLS outputs	2,900	
HC (L1)	Benign vs. malignant	ISIC 2018 classification (HAM10000-style) [15]	7,000	
HC (multi-class)	Extended classification	ISIC 2016/2017/2019/2020/2024, BCN20000, MiLK-10K [14], [16], [17]	$\approx$ 100,000 im	
MedProc	Clinical NLP	MIMIC-III, MIMIC-IV Note [24], [25]	50,000	
IT-Fusion	Multimodal fusion	Derm1M + HAM10000 + PAD-UFES-20 + ISIC 2016–2024 + BCN20000 + MIMIC text [14]–[16], [20]	700,000	10
VQA	Med & derm VQA	DermaVQA, SkinCancerVQA, SkinQA, PathVQA, VQA-RAD, HEALMQA [34]–[37]	80,000	1

Inference will be orchestrated via FastAPI, with GPU-accelerated models containerized (e.g., Docker) and exposed via internal APIs.

VII. Evaluation Plan

A. Stage-Specific Metrics

We define metrics per stage:

- **SLS:** Dice coefficient, Intersection-over-Union (IoU), and boundary F1 scores on ISIC segmentation benchmarks [14], [46].
- **SLRC:** Accuracy of lesion coverage (percentage of lesion pixels inside ROI) and preservation of border areas.
- **HC:** Accuracy, balanced accuracy, AUC-ROC, F1-score for benign vs. malignant and multi-class tasks [15]–[17].
- **ABCD/ABCDE:** Correlation and mean absolute error between computed ABCDE values and dermatologist-provided scores where available; agreement on risk bands used in ABCD-based scoring systems [1]–[4].
- **MedProc:** Precision, recall, F1 for clinical entities and attributes (e.g., itching, bleeding, duration).
- **IT-Fusion:** Calibration metrics (Brier score, expected calibration error), risk-curve analyses, and overall AUC-ROC on malignant vs. benign classification with and without text features [21], [22].
- **VQA:** BLEU, ROUGE-L, and answer exact match metrics, plus specialized metrics used in HEALMQA and Med-VQA challenges, alongside human evaluation by dermatologists for correctness and helpfulness [34]–[37], [55].

B. End-to-End Evaluation

End-to-end evaluation includes:

- Measuring improvement in malignant lesion recall when integrating MedProc and IT-Fusion compared to image-only baselines.
- Assessing answer faithfulness by verifying whether VQA rationales are consistent with ABCDE and fusion-stage JSON.
- Conducting small-scale user studies with dermatologists to assess usability, explanation quality, and potential for workflow integration, informed by AMIE and dermatology chatbot studies [40]–[44].

VIII. Ethical, Safety, and Deployment Considerations

Deploying NutriDermAI in clinical or patient-facing settings requires robust safeguards:

- **Scope and disclaimers:** The system is positioned as decision support rather than a standalone diagnostic tool.
- **Escalation and uncertainty handling:** When confidence is low or high-risk patterns are detected, the system explicitly recommends dermatologist consultation and avoids overconfident statements.
- **Privacy:** Image and text data are processed under appropriate anonymization and secure storage guidelines.
- **Bias and fairness:** Evaluation on diverse datasets and continual monitoring for performance disparities across skin tones and demographic subgroups.

These choices are guided by recent reviews of medical MM-LLMs, ChatGPT in dermatology, and large-scale conversational diagnostic systems [33], [42], [43].

IX. Conclusion and Future Work

This paper proposes NutriDermAI, a multimodal dermatology assistant that unifies lesion segmentation, ROI cropping, hierarchical classification, ABCDE-aware feature computation, clinical text understanding, multimodal fusion, and interactive VQA within a single explainable pipeline. Building on dermatology foundation models, clinical NLP, and medical VQA literature, NutriDermAI aims to provide clinically meaningful, interpretable, and conversational support for skin lesion assessment.

Future work includes:

- Collecting dermatologist-annotated ABCDE scores and longitudinal image sequences for more robust Evolution modeling.
- Exploring parameter-efficient finetuning of large multimodal models to fit resource-constrained deployments.
- Extending the architecture to additional visual dermatology tasks (e.g., whole-body screening) and other specialties such as radiology and ophthalmology.
- Conducting prospective clinical trials to quantify impact on diagnostic accuracy, workflow efficiency, and patient experience.

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